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SEMICONDUCTOR STRUCTURE

Abstract

A semiconductor structure includes an insulator, a semiconductor fin, a gate stack, a gate contact, a source/drain material, and a source/drain contact structure. The semiconductor fin protrudes from the insulator. The gate stack is disposed on the semiconductor fin and the insulator. The gate contact is disposed on and electrically connected to the gate stack. The source/drain material is disposed on the semiconductor fin. The source/drain contact structure is disposed on and electrically connected to the source/drain material. The semiconductor fin extends along a first direction, the gate stack extends along a second direction different from the first direction. An offset S in the second direction between the gate contact and the source/drain contact structure satisfies: $0 < S \leq (W/2 + D/2)$, wherein W is a width of the semiconductor fin, and D is a dimension of the gate contact.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of and claims the priority benefit of a prior application Ser. No. 17/727,846, filed on Apr. 25, 2022, and now allowed. The entirety of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

[0002] With advances in semiconductor technology, there has been increasing demand for higher storage capacity, faster processing systems, higher performance, and lower costs. To meet these demands, the semiconductor industry continues to scale down the dimensions of semiconductor devices, such as metal oxide semiconductor field effect transistors (MOSFETs), including planar MOSFETs and finFETs.

[0003] Such scaling down has increased the complexity of semiconductor manufacturing processes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIGS. 1, 2 and 3A are three-dimensional views of intermediate stages in the manufacturing of the semiconductor structure, in accordance with some embodiments. FIGS. 3B and 3C are cross-sectional views of the intermediate stage shown in FIG. 3A, in accordance with some embodiments.

[0006] FIGS. 4A through 10B are cross-sectional views of further intermediate stages in the manufacturing of the semiconductor structure, in accordance with some embodiments.

[0007] FIGS. 11-15 are schematic top views of semiconductor structures, in accordance with some embodiments.

DETAILED DESCRIPTION

[0008] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments

and/or configurations discussed.

[0009] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0010] The fins may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins.

[0011] FIGS. 1 through 10B are various views of intermediate stages in the manufacturing of the semiconductor structure, e.g., FinFETs, in accordance with some embodiments. FIGS. 1, 2 and 3A are three-dimensional views. As shown in FIG. 3A, cross-section A-A is along a longitudinal axis of the semiconductor fin 52A between the source/drain materials 70A and 70B, and cross-section B-B is perpendicular to cross-section A-A and extends through the source/drain materials 70A and 70B. FIGS. 3B, 4A, 6A, 7A, 8A, 9A, and 10A are cross-sectional views illustrated along reference cross-section A-A in FIG. 3A. FIGS. 3C, 4B, 5B, 6B, 7B, 8B, 9B, and 10B are cross-sectional views illustrated along reference cross-section B-B in FIG. 3A.

[0012] In FIG. 1, a substrate 50 is provided. The substrate 50 may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The substrate 50 may be a wafer, such as a silicon wafer. Generally, an SOI substrate is a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a buried oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, typically a silicon or glass substrate. Other substrates, such as a multi-layered or gradient substrate may also be used. In some embodiments, the semiconductor material of the substrate 50 may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof.

[0013] The substrate 50 has a region 50A and a region 50B. The region 50A can be for forming n-type devices, such as NMOS transistors, e.g., n-type FinFETs. The region 50B can be for forming p-type devices, such as PMOS transistors, e.g., p-type FinFETs. However, the types of the devices are not limited in the present disclosure. The region 50A may be physically separated from the region 50B, and any number of device features (e.g., other active devices, doped regions, isolation structures, etc.) may be disposed between the region 50A and the region 50B. For brevity, only region 50A is shown in FIGS. 3B through 10B. However, similar process may also be performed in region 50B.

[0014] Fins 52A and 52B are formed extending from the substrate 50. The fins 52A and 52B are semiconductor strips. In some embodiments, the fins 52A and 52B may be formed in the substrate 50 by etching trenches in the substrate 50. The etching may be any acceptable etch process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etch may be anisotropic. After formations, the fins 52A and 52B have a width W. The width W can be in

the range of about 3 nm to about 30 nm.

[0015] The fins may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins.

[0016] STI regions (insulator) **56** are formed over the substrate **50** and between neighboring fins, e.g., fins **52A** and **52B**. As an example to form the STI regions **56**, an insulation material is formed over the intermediate structure. The insulation material may be an oxide, such as silicon oxide, a nitride, the like, or a combination thereof, and may be formed by a high density plasma chemical vapor deposition (HDP-CVD), a flowable chemical vapor deposition (FCVD) (e.g., a chemical vapor deposition (CVD) based material deposition in a remote plasma system and post curing to make it convert to another material, such as an oxide), the like, or a combination thereof. Other insulation materials formed by any acceptable process may be used. In the illustrated embodiment, the insulation material is silicon oxide formed by a FCVD process. An anneal process may be performed once the insulation material is formed. In an embodiment, the insulation material is formed such that excess insulation material covers the fins **52A** and **52B**. Some embodiments may utilize multiple layers. For example, in some embodiments a liner (not shown) may first be formed along a surface of the substrate **50** and the fins **52A** and **52B**. Thereafter, a fill material, such as those discussed above may be formed over the liner. A removal process is applied to the insulation material to remove excess insulation material over the fins **52A** and **52B**. In some embodiments, a planarization process such as a chemical mechanical polish (CMP), an etch back process, combinations thereof, or the like may be utilized. The planarization process exposes the fins **52A** and **52B** such that top surfaces of the fins **52A** and **52B** and the insulation material are level after the planarization process is complete. The insulation material is then recessed, with remaining portions of the insulation material forming the STI regions **56**. The insulation material is recessed such that upper portions of fins **52A** and **52B** in the region **50A** and in the region **50B** protrude from between neighboring STI regions **56**. After the recessing, exposed portions of the fins **52A** and **52B** extend a height *H* above top surfaces of the STI regions **56**. In some embodiments, the height *H* can be greater than about 40 nm, such as in the range of about 50 nm to about 80 nm. In some embodiments, the height *H* can be in the range of about 15 nm to about 50 nm. The exposed portions of the fins **52A** and **52B** include what will be channel regions of the resulting FinFETs. The process described above is just one example of how the fins **52A** and **52B** may be formed and is not intended to be limiting.

[0017] Further, appropriate wells (not shown) may be formed in the fins **52A** and **52B** and/or the substrate **50**. In some embodiments, a P well may be formed in the region **50A**, and an N well may be formed in the region **50B**. In some embodiments, a P well or an N well are formed in both the region **50A** and the region **50B**. In the embodiments with different well types, the different implant steps for the region **50A** and the region **50B** may be achieved using a photoresist or other masks (not shown). The n-type impurities may be phosphorus, arsenic, antimony, or the like implanted in the region to a concentration of equal to or less than 10^{18} cm^{-3} , such as between about 10^{17} cm^{-3} and about 10^{18} cm^{-3} . The p-type impurities may be boron, BF₂, indium, or the like implanted in the region to a concentration of equal to or less than 10^{18} cm^{-3} , such as between about 10^{17} cm^{-3} and about 10^{18} cm^{-3} .

[0018] In FIG. 2, dummy gate dielectrics **60** are formed over the fins **52A** and **52B** and dummy gates **62** are formed over the dummy gate dielectrics **60**. The dummy gate dielectrics **60** and

dummy gates **62** may be collectively referred to as “dummy gate stacks,” with each dummy gate stack including a dummy gate dielectric **60** and a dummy gate **62**. The dummy gate stacks extend along sidewalls of the fins **52A** and **52B**. Although only one dummy gate stack is illustrated in FIG. **2**, it should be appreciated that multiple dummy gate stacks are simultaneously formed, and each fin may have multiple dummy gate stacks formed thereon (as shown in FIG. **3B**).

[0019] As an example of forming the dummy gate dielectrics **60** and dummy gates **62**, a dummy dielectric layer is formed on the fins **52A** and **52B**. The dummy dielectric layer may be, for example, silicon oxide, silicon nitride, a combination thereof, or the like, and may be deposited or thermally grown according to acceptable techniques. A dummy gate layer is formed over the dummy dielectric layer, and a mask layer is formed over the dummy gate layer. The dummy gate layer may be deposited over the dummy dielectric layer and then planarized, such as by a CMP. The mask layer may be deposited over the dummy gate layer. The dummy gate layer may be a conductive or non-conductive material and may be selected from a group including amorphous silicon, polycrystalline-silicon (polysilicon), poly-crystalline silicon germanium (poly-SiGe), metallic nitrides, metallic silicides, metallic oxides, and metals. The dummy gate layer may be deposited by physical vapor deposition (PVD), CVD, sputter deposition, or other techniques known and used in the art for depositing conductive materials. The dummy gate layer may be made of other materials that have a high etching selectivity from the etching of isolation regions. The mask layer may include, for example, silicon nitride, silicon oxynitride, or the like. In this example, a single dummy gate layer and a single mask layer are formed across the region **50A** and the region **50B**. In some embodiments, the dummy dielectric layer may be deposited such that the dummy dielectric layer covers the STI regions **56**, extending between the dummy gate layer and the STI regions **56**. The mask layer is then patterned using acceptable photolithography and etching techniques to form masks **64**. The pattern of the masks **64** is then transferred to the dummy gate layer by an acceptable etching technique to form dummy gates **62**. The pattern of the masks **64** is further transferred to the dummy dielectric layer to form dummy gate dielectrics **60**. The dummy gates **62** cover respective channel regions **58** of the fins **52A** and **52B** (as can be seen in FIG. **3B**). The pattern of the masks **64** may be used to physically separate each of the dummy gates **62** from adjacent dummy gates. The dummy gates **62** may also have a lengthwise direction substantially perpendicular to the lengthwise direction of respective fins **52A** and **52B**.

[0020] In FIGS. **3A** through **3C**, gate spacers **66** are formed on exposed surfaces of the dummy gates **62**, the masks **64**, and/or the fins **52A** and **52B**. The gate spacers **66** may be formed by conformally depositing an insulating material and subsequently etching the insulating material. The insulating material of the gate spacers **66** may be silicon nitride, silicon oxide, silicon carbonitride, silicon oxycarbonitride, a combination thereof, or the like. In some embodiments (not shown), the gate spacers **66** are formed from a multi-layered insulating material, and include multiple layers. For example, the gate spacers **66** may include a layer of silicon oxide disposed between two layers of silicon nitride, or may include multiple layers of silicon oxycarbonitride.

[0021] Epitaxial source/drain materials **70A** and **70B** are then formed on the fins **52A** and **52B** to exert stress in the channel regions **58**, thereby improving performance. The epitaxial source/drain materials **70A** and **70B** are formed on the fins **52A** and **52B** such that each dummy gate **62** is disposed between respective neighboring pairs of the epitaxial source/drain materials **70A** and **70B**. In some embodiments the epitaxial source/drain materials **70A** and **70B** may extend into, and may also penetrate through the fins **52A** and **52B**. In some embodiments, the gate spacers **66** are used to separate the epitaxial source/drain materials **70A** and **70B** from the dummy gates **62** by an appropriate lateral distance so that the epitaxial source/drain materials **70A** and **70B** do not short out subsequently formed gates of the resulting FinFETs.

[0022] The epitaxial source/drain materials **70A** and **70B** may be formed by etching source/drain regions of the fins **52A** and **52B** to form recesses **52R** in the fins **52A** and **52B**. Then, the epitaxial source/drain materials **70A** and **70B** are epitaxially grown in the recesses **52R**. In some

embodiments, the epitaxial source/drain materials **70A** and **70B** may include any acceptable material, such as appropriate for n-type FinFETs. For example, if the fins **52A** and **52B** are silicon, the epitaxial source/drain materials **70A** and **70B** may include materials exerting a tensile strain in the channel regions **58**, such as silicon, silicon carbide, phosphorous doped silicon carbide, silicon phosphide, or the like. In some other embodiments, the epitaxial source/drain materials **70A** and **70B** may include any acceptable material, such as appropriate for p-type FinFETs. For example, if the fins **52A** and **52B** are silicon, the epitaxial source/drain materials **70A** and **70B** may comprise materials exerting a compressive strain in the channel regions **58**, such as silicon germanium, germanium, germanium tin, or the like. The epitaxial source/drain materials **70A** and **70B** may have surfaces raised from respective surfaces of the fins **52A** and **52B** and may have facets. In some embodiments, as illustrated by FIG. 3A, adjacent epitaxial source/drain materials **70A** and **70B** remain separated after the epitaxy process is completed.

[0023] The epitaxial source/drain materials **70A** and **70B** and/or the fins **52A** and **52B** may be doped with impurities to form source/drain regions. The source/drain regions may have an impurity concentration of between about 10^{19} cm.⁻³ and about 10^{21} cm.⁻³. The n-type and/or p-type impurities for source/drain regions may be any of the impurities previously discussed. The epitaxial source/drain materials **70A** and **70B** are in situ doped during growth. Doping during growth instead of by implantation can avoid stress relaxation in the channel regions **58** that occurs during dopant implantation. Performance of the resulting FinFETs may thus be improved.

[0024] In FIGS. 4A and 4B, a first ILD layer **102** is deposited over the intermediate structure. The first ILD layer **102** may be formed of a dielectric material, and may be deposited by any suitable method, such as CVD, plasma-enhanced CVD (PECVD), or FCVD. Dielectric materials may include Phospho-Silicate Glass (PSG), Boro-Silicate Glass (BSG), Boron-Doped Phospho-Silicate Glass (BPSG), undoped Silicate Glass (USG), or the like. Other insulation materials formed by any acceptable process may be used. In some embodiments, a contact etch stop layer (CESL) (not shown) is disposed between the first ILD layer **102** and the epitaxial source/drain materials **70A** and **70B**, the masks **64**, and the gate spacers **66**. The CESL may comprise a dielectric material, such as, silicon nitride, silicon oxide, silicon oxynitride, or the like, having a different etch rate than the material of the first ILD layer **102**.

[0025] In FIGS. 5A and 5B, a planarization process, such as a CMP, may be performed to level the top surface of the first ILD layer **102** with the top surfaces of the dummy gates **62** or the masks **64**. The planarization process may also remove the masks **64** on the dummy gates **62**, and portions of the gate spacers **66** along sidewalls of the masks **64**. After the planarization process, top surfaces of the dummy gates **62**, the gate spacers **66**, and the first ILD layer **102** are level. Accordingly, the top surfaces of the dummy gates **62** are exposed through the first ILD layer **102**. In some embodiments, the masks **64** may remain, in which case the planarization process levels the top surface of the first ILD layer **102** with the top surfaces of the masks **64**.

[0026] In FIGS. 6A and 6B, the dummy gates **62** are removed and are replaced with metal gates **110**. The metal gates **110** include gate dielectrics **112** and gate electrodes **114**. As an example to form the metal gates **110**, the dummy gates **62**, and the masks **64** if present, are removed in one or more etching step(s), so that recesses are formed. Portions of the dummy gate dielectrics **60** in the recesses may also be removed. In some embodiments, the dummy gates **62** are removed by an anisotropic dry etch process. For example, the etching process may include a dry etch process using reaction gas(es) that selectively etch the dummy gates **62** without etching the first ILD layer **102** or the gate spacers **66**. The recesses expose the fin **52A**. Specifically, the channel regions **58** are exposed by the recesses. Channel region **58** is disposed between neighboring pairs of the epitaxial source/drain materials **70A** and **70B**. During the removal, the dummy gate dielectrics **60** may be used as etch stop layers when the dummy gates **62** are etched. The dummy gate dielectrics **60** may then be optionally removed after the removal of the dummy gates **62**.

[0027] After the removal, the gate dielectrics **112** are deposited conformally in the recesses, such as on the top surfaces and the sidewalls of the fin **52A** and on sidewalls of the gate spacers **66**. The gate dielectrics **112** may also be formed on top surface of the first ILD layer **102**. In accordance with some embodiments, the gate dielectrics **112** comprise silicon oxide, silicon nitride, or multilayers thereof. In some embodiments, the gate dielectrics **112** include a high-k dielectric material, and in these embodiments, the gate dielectrics **112** may have a k value greater than about 7.0, and may include a metal oxide or a silicate of Hf, Al, Zr, La, Mg, Ba, Ti, Pb, and combinations thereof. The formation methods of the gate dielectrics **112** may include Molecular-Beam Deposition (MBD), atomic layer deposition (ALD), PECVD, and the like. The gate electrodes **114** are deposited over the gate dielectrics **112**, respectively, and fill the remaining portions of the recesses. The gate electrodes **114** may include a metal-containing material such as TiN, TiO, TaN, TaC, Co, Ru, Al, W, combinations thereof, or multi-layers thereof. For example, although a single-layered gate electrode **114** is illustrated, each gate electrode **114** may comprise any number of liner layers, any number of work function tuning layers, and a fill material. After the filling of the gate electrodes **114**, a planarization process, such as a CMP, may be performed to remove the excess portions of the gate dielectrics **112** and the material of the gate electrodes **114**, which excess portions are over the top surface of the first ILD layer **102**. The remaining portions of material of the gate electrodes **114** and the gate dielectrics **112** thus form replacement gates of the resulting FinFETs. The metal gates **110** may be also referred to as “gate stacks” or “replacement gate stacks.” The metal gates **110** may extend along sidewalls of the channel regions **58** of the fin **52A**.

[0028] The formation of the gate dielectrics **112** in the region **50A** and the region **50B** may occur simultaneously such that the gate dielectrics **112** in each region are formed from the same materials, and the formation of the gate electrodes **114** may occur simultaneously such that the gate electrodes **114** in each region are formed from the same materials. In some embodiments, the gate dielectrics **112** in each region may be formed by distinct processes, such that the gate dielectrics **112** may be different materials, and/or the gate electrodes **114** in each region may be formed by distinct processes, such that the gate electrodes **114** may be different materials.

[0029] In FIGS. 7A and 7B, contact openings **120** are formed through the first ILD layer **102** and the CESL, exposing the epitaxial source/drain materials **70A** and **70B**. The contact openings **120** may be formed using acceptable photolithography and etching techniques.

[0030] In FIGS. 8A and 8B, lower source/drain contacts **124A** and **124B** are formed in the contact openings **120**. A liner, such as a diffusion barrier layer, an adhesion layer, or the like, and a conductive material are formed in the contact openings **120**. The liner may include titanium, titanium nitride, tantalum, tantalum nitride, or the like. The conductive material may be copper, a copper alloy, silver, gold, tungsten, cobalt, aluminum, nickel, or the like. A planarization process, such as a CMP, may be performed to remove excess material from a surface of the first ILD layer **102**. The remaining liner and conductive material form the lower source/drain contacts **124A** and **124B** in the contact openings **120**. The lower source/drain contacts **124A** and **124B** are physically and electrically coupled to the epitaxial source/drain materials **70A** and **70B**.

[0031] In FIGS. 9A and 9B, a second ILD layer **130** is deposited over the first ILD layer **102** and lower source/drain contacts **124A** and **124B**. In some embodiment, the second ILD layer **130** is a flowable film formed by a flowable CVD method. In some embodiments, the second ILD layer **130** is formed of a dielectric material such as PSG, BSG, BPSG, USG, or the like, and may be deposited by any suitable method, such as CVD and PECVD.

[0032] In FIGS. 10A and 10B, gate contacts **134** and upper source/drain contacts **136A** and **136B** are formed through the second ILD layer **130**. Openings for the gate contacts **134** and upper source/drain contacts **136A** and **136B** are formed through the second ILD layer **130**. The openings may be formed using acceptable photolithography and etching techniques. A liner, such as a diffusion barrier layer, an adhesion layer, or the like, and a conductive material are formed in the openings. The liner may include titanium, titanium nitride, tantalum, tantalum nitride, or the like.

The conductive material may be copper, a copper alloy, silver, gold, tungsten, cobalt, aluminum, nickel, or the like. A planarization process, such as a CMP, may be performed to remove excess material from a surface of the second ILD layer **130**. The remaining liner and conductive material form the gate contacts **134** and upper source/drain contacts **136A** and **136B** in the openings. The upper source/drain contacts **136A** and **136B** are physically and electrically coupled to the lower source/drain contacts **124A** and **124B**, respectively, and the gate contacts **134** are physically and electrically coupled to the metal gates **110**. The gate contacts **134** and upper source/drain contacts **136A** and **136B** may be formed in different processes, or may be formed in the same process.

[0033] FIG. **11** illustrates a schematic top view of the semiconductor structure in FIGS. **10A** and **10B**. In FIG. **11**, some features of the FinFETs, such as the source/drain materials **70A** and **70B**, are omitted for illustration clarity. Referring to FIGS. **10A**, **10B**, and **11**, the semiconductor structure may include an insulator **56**, a first semiconductor fin **52A**, a gate stack **110**, a first source/drain material **70A**, and a second source/drain material **70B**. The first semiconductor fin **52A** protrudes from the insulator **56**. The gate stack **110** is disposed on the first semiconductor fin **52A** and the insulator **56**. The gate stack **110** may cover the insulator **56** and an intersect region IR of the first semiconductor fin **52A**. In some embodiments, the gate stack **110** wraps the intersect region IR of the first semiconductor fin **52A**. The first semiconductor fin **52A** substantially extends along a first direction D1, and the gate stack **110** substantially extends along a second direction D2 different from the first direction D1. In some embodiments, the first direction D1 is perpendicular to the second direction D2. The first source/drain material **70A** and the second source/drain material **70B** may be disposed on the first semiconductor fin **52A** and on opposite sides of the intersect region IR of the first semiconductor fin **52A**.

[0034] The semiconductor structure may further include a gate contact **134**, a first source/drain contact structure **125A**, and a second source/drain contact structure **125B**. The gate contact **134** is disposed on and electrically connected to the gate stack **110**. In some embodiments, a ratio between the dimension of the gate contact **134** and a width of the gate stack **110** is from about 2.4 to about 3.2. The first source/drain contact structure **125A** may be disposed on and electrically connected to the first source/drain material **70A**, and the second source/drain contact structure **125B** may be disposed on and electrically connected to the second source/drain material **70B**. The first source/drain contact structure **125A** and the second source/drain contact structure **125B** are disposed on opposite sides of the intersect region IR of the first semiconductor fin **52A**. The first source/drain contact structure **125A** and the second source/drain contact structure **125B** may be symmetry with respect to the gate stack **110**.

[0035] Specifically, the first source/drain contact structure **125A** may include a lower source/drain contact **124A** and an upper source/drain contact **136B**, and the second source/drain contact structure **125B** may include a lower source/drain contact **124B** and an upper source/drain contact **136B**. The upper source/drain contact **136A** may be disposed on and electrically connected to the lower source/drain contact **124A**, and the upper source/drain contact **136B** may be disposed on and electrically connected to the lower source/drain contact **124B**. In some embodiments, as shown in FIG. **11**, the upper source/drain contact **136A** may not overlap with the gate contact **134** in the first direction D1, and the upper source/drain contact **136B** may not overlap with the gate contact **134** in the first direction D1, either.

[0036] Referring to FIG. **11**, an offset S in the second direction D2 may exist between the gate contact **134** and a source/drain contact structure, e.g., the first source/drain contact structure **125A**. As shown in FIG. **11**, the offset S is determined by the offset in the second direction D2 between the center of the gate contact **134** and the center of the first source/drain contact structure **125A**. Under another aspect, the offset S may be considered as an offset in a lengthwise direction of the gate stack **110** between the gate contact **134** and the first source/drain contact structure **125A**. Under yet another aspect, the offset S may be considered as an offset in a widthwise direction of the first semiconductor fin **52A** between the gate contact **134** and the first source/drain contact

structure **125A**. In some embodiments, the offset S satisfied: $0 < S \leq (W/2 + D/2)$, wherein W is a width of the first semiconductor fin **52A**, and D is a dimension of the gate contact **134**. Such relative configuration between the gate contact and the source/drain contact structure may increase design flexibility and reduce occupied area of the device, while the performance of the semiconductor device, e.g., the resulting FinFETs, may be more predictable and more reliable. [0037] As shown in FIG. **11**, the gate contact **134** and the first semiconductor fin **52A** may overlap in a stacking direction $D3$ of the semiconductor structure. Under another aspect, the gate contact **134** and the intersect region IR of the first semiconductor fin **52A** may overlap in a direction perpendicular to the lengthwise direction of the gate stack **110** and a lengthwise direction of the first semiconductor fin **52A**. As shown in FIG. **11**, the gate contact **134** and the insulator **56** may overlap in a stacking direction $D3$ of the semiconductor structure as well.

[0038] In some embodiments, the semiconductor structure may further include a second semiconductor fin **52B** protruding from the insulator **56**. The second semiconductor fin **52B** may substantially extend along the first direction $D1$, and may be of similar width and similar height as the first semiconductor fin **52A**. The gate stack **110** may be disposed on both the first semiconductor fin **52A** and the second semiconductor fin **52B**. In some embodiments, an interval $V1$ between the first semiconductor fin **52A** and the second semiconductor fin **52B**, which can be determined by a shortest distance between the first semiconductor fin **52A** and the second semiconductor fin **52B**, may be from about $0.07\ \mu\text{m}$ to about $0.08\ \mu\text{m}$.

[0039] FIGS. **12-15** illustrate schematic top views of semiconductor structures that are similar to the semiconductor structure illustrated in FIGS. **10A**, **10B**, and **11**, with like elements labeled with like numerical references. In some embodiments, the semiconductor structures of FIGS. **12-15** may be formed using similar materials and methods as the semiconductor structure of FIGS. **10A**, **10B**, and **11**, described above with reference to FIGS. **1-10B**, and the description is not repeated herein for the sake of brevity.

[0040] Referring to FIG. **12**, a first end $E1$ of the gate contact **134** is aligned with an edge $E2$ of the first semiconductor fin **52A** in the stacking direction $D3$ of the semiconductor structure, wherein the first end $E1$ is an end of the gate contact **134** having a minimum distance between the end $E1$ and a central axis $C1$ of the first semiconductor fin **52A**. Under another aspect, an interval in the lengthwise direction of the gate stack **110** between the first end $E1$ of the gate contact **134** and an edge $E2$ of the first semiconductor fin **52A** is zero. Referring to FIG. **13**, in some embodiments, the gate contact **134** and the insulator **56** may not overlap in the stacking direction $D3$ of the semiconductor structure.

[0041] Referring to FIG. **14**, in some embodiments, the semiconductor structure may include a first gate contact **134A** similar to the gate contact **134** shown in FIG. **11**, and may further include a second gate contact **134B** disposed on and electrically connected to the gate stack **110**. In some embodiments, the second gate contact **134B** and the intersect region IR of the first semiconductor fin **52A** may overlap in a direction perpendicular to the lengthwise direction of the gate stack **110** and a lengthwise direction of the first semiconductor fin **52A**, e.g., in the direction $D3$ shown in FIG. **14**. In some embodiments, the first gate contact **134A** and the second gate contact **134B** may be symmetrical with respect to a central axis $C1$ of the first semiconductor fin **52A**. The symmetrical arrangement of the first gate contact **134A** and the second gate contact **134B** may provide a semiconductor device with more favorable and controllable feature.

[0042] Referring to FIG. **15**, in some embodiments, the second gate contact **134B** and the intersect region IR of the first semiconductor fin **52A** may not overlap in a direction perpendicular to the lengthwise direction of the gate stack **110** and a lengthwise direction of the first semiconductor fin **52A**, e.g., in the direction $D3$ shown in FIG. **15**. In some embodiments, the first gate contact **134A** and the second gate contact **134B** may be asymmetrical with respect to a central axis $C1$ of the first semiconductor fin **52A**.

[0043] In the above-mentioned semiconductor structure, with the relative configuration between

the gate contact and the source/drain contact structure fulfilling certain conditions, the design flexibility of the devices may be increased and the occupied area of the device may be reduced, while the performance of the semiconductor device may be more predictable and more reliable.

[0044] In accordance with some embodiments of the present disclosure, a semiconductor structure includes an insulator, a semiconductor fin, a gate stack, a gate contact, a source/drain material, and a source/drain contact structure. The semiconductor fin protrudes from the insulator. The gate stack is disposed on the semiconductor fin and the insulator. The gate contact is disposed on and electrically connected to the gate stack. The source/drain material is disposed on the semiconductor fin. The source/drain contact structure is disposed on and electrically connected to the source/drain material. The semiconductor fin extends along a first direction, the gate stack extends along a second direction different from the first direction. An offset S in the second direction between the gate contact and the source/drain contact structure satisfies: $0 < S \leq (W/2 + D/2)$, wherein W is a width of the semiconductor fin, and D is a dimension of the gate contact. In some embodiments, the first direction is perpendicular to the second direction. In some embodiments, the gate contact and the semiconductor fin overlap in a stacking direction of the semiconductor structure. In some embodiments, the gate contact and the insulator do not overlap in a stacking direction of the semiconductor structure. In some embodiments, the gate contact and the insulator overlap in a stacking direction of the semiconductor structure. In some embodiments, the gate contact includes a first end having a minimum distance between the first end and a central axis of the semiconductor fin, and the first end of the gate contact is aligned with an edge of the semiconductor fin in a stacking direction of the semiconductor structure. In some embodiments, the source/drain contact structure includes a lower source/drain contact and an upper source/drain contact electrically connected to the lower source/drain contact. In some embodiments, the upper source/drain contact is not overlapped with the gate contact in the first direction.

[0045] In accordance with some embodiments of the present disclosure, a semiconductor structure includes an insulator, a first semiconductor fin, a gate stack, a first gate contact, a source/drain material, and a source/drain contact structure. The first semiconductor fin protrudes from the insulator. The gate stack covers the insulator and an intersect region of the first semiconductor fin. The first gate contact is disposed on and electrically connected to the gate stack. The source/drain material is disposed on the first semiconductor fin. The source/drain contact structure is disposed on and electrically connected to the source/drain material. An offset S in a lengthwise direction of the gate stack between the first gate contact and the source/drain contact structure satisfies: $0 < S \leq (W/2 + D/2)$, wherein W is a width of the first semiconductor fin, and D is a dimension of the first gate contact. In some embodiments, the first gate contact and the intersect region of the first semiconductor fin overlap in a direction perpendicular to the lengthwise direction of the gate stack and a lengthwise direction of the first semiconductor fin. In some embodiments, the first gate contact includes a first end having a minimum distance between the first end and a central axis of the first semiconductor fin, and an interval in the lengthwise direction of the gate stack between the first end of the first gate contact and an edge of the first semiconductor fin is zero. In some embodiments, the semiconductor structure further includes a second semiconductor fin protruding from the insulator, wherein the gate stack is disposed on the second semiconductor fin. In some embodiments, an interval between the first semiconductor fin and the second semiconductor fin is from about $0.07\ \mu\text{m}$ to about $0.08\ \mu\text{m}$. In some embodiments, the semiconductor structure further includes a second gate contact disposed on and electrically connected to the gate stack. In some embodiments, the second gate contact and the intersect region of the first semiconductor fin overlap in a direction perpendicular to the lengthwise direction of the gate stack and a lengthwise direction of the first semiconductor fin. In some embodiments, the first gate contact and the second gate contact are symmetry with respect to a central axis of the first semiconductor fin.

[0046] In accordance with some embodiments of the present disclosure, a semiconductor structure includes a semiconductor fin, a gate stack, a gate contact, a first source/drain material and a second

source/drain material, and a first source/drain contact structure. The gate stack wraps an intersect region of the semiconductor fin. The gate contact is disposed on and electrically connected to the gate stack. The first source/drain material and the second source/drain material are disposed on the semiconductor fin and on opposite sides of the intersect region of the semiconductor fin. The first source/drain contact structure is disposed on and electrically connected to the first source/drain material. An offset S between the gate contact and the first source/drain contact structure in a widthwise direction of the semiconductor fin satisfies: $0 < S \leq (W/2 + D/2)$, wherein W is a half of a width of the semiconductor fin, and D is a half of a dimension of the gate contact. In some embodiments, the semiconductor structure further includes a second source/drain contact structure disposed on and electrically connected to the second source/drain material, wherein the first source/drain contact structure and the second source/drain contact structure are disposed on opposite sides of the intersect region of the semiconductor fin. In some embodiments, the first source/drain contact structure and the second source/drain contact structure are symmetry with respect to the gate stack. In some embodiments, a ratio between the dimension of the gate contact and a width of the gate stack is from about 2.4 to about 3.2.

[0047] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure, comprising: a semiconductor channel region; a gate stack covering and intersected with the semiconductor channel region; a gate contact disposed on and electrically connected to the gate stack; a source/drain material disposed on the semiconductor channel region; and a source/drain contact structure disposed on and electrically connected to the source/drain material; wherein the semiconductor channel region extends along a first direction, the gate stack extends along a second direction different from the first direction, and an offset S in the second direction between the gate contact and the source/drain contact structure satisfied: $0 < S \leq (W/2 + D/2)$, wherein W is a width of the semiconductor fin, and D is a dimension of the gate contact.
2. The semiconductor structure of claim 1, wherein the first direction is perpendicular to the second direction.
3. The semiconductor structure of claim 1, wherein the gate contact overlaps with the semiconductor channel region in a third direction perpendicular to the first direction and the second direction.
4. The semiconductor structure of claim 1 further comprising an insulator, wherein the gate contact does not overlap with the insulator in a third direction perpendicular to the first direction and the second direction.
5. The semiconductor structure of claim 1 further comprising an insulator, wherein the gate contact overlaps with the insulator in a third direction perpendicular to the first direction and the second direction.
6. The semiconductor structure of claim 1, wherein the gate contact includes a first end having a minimum distance between the first end and a central axis of the semiconductor channel region, and the first end of the gate contact is aligned with an edge of the semiconductor channel region in a third direction perpendicular to the first direction and the second direction.
7. The semiconductor structure of claim 1, wherein the source/drain contact structure comprises a

lower source/drain contact and an upper source/drain contact, and the lower source/drain contact is between the source/drain material and the upper source/drain contact.

8. The semiconductor structure of claim 7, wherein the upper source/drain contact is not overlapped with the gate contact.

9. A semiconductor structure, comprising: a field effect transistor (FET) comprising a first channel region, a gate stack intersected with the first channel region, and a source/drain material; a first gate contact disposed on and electrically connected to the gate stack of the FET; and a source/drain contact structure disposed on and electrically connected to the source/drain material of the FET; wherein an offset in a lengthwise direction of the gate stack is between the first gate contact and the source/drain contact structure, the offset is greater than zero, and the offset is not greater than an average value of a width of the channel region and a dimension of the gate contact.

10. The semiconductor structure of claim 9, wherein the first gate contact overlaps with the intersect region of the first channel region.

11. The semiconductor structure of claim 9, wherein the first gate contact includes a first end having a minimum distance between the first end and a central axis of the first channel region, and an interval in the lengthwise direction of the gate stack between the first end of the first gate contact and an edge of the first channel region is zero.

12. The semiconductor structure of claim 9, further comprising a second channel region, wherein the gate stack intersects the second channel region.

13. The semiconductor structure of claim 12, wherein an interval between the first channel region and the second channel region is from about 0.07 μm to about 0.08 μm .

14. The semiconductor structure of claim 9, further comprising a second gate contact disposed on and electrically connected to the gate stack.

15. The semiconductor structure of claim 14, wherein the second gate contact overlaps with the intersect region of the first channel region.

16. The semiconductor structure of claim 14, wherein the first gate contact is symmetrical with the second gate contact with respect to a central axis of the first channel region.

17. A semiconductor structure, comprising: a channel region; a gate stack wrapping an intersect region of the channel region; a gate contact disposed on and electrically connected to the gate stack; a first source/drain material and a second source/drain material disposed on opposite sides of the intersect region of the channel region; and a first source/drain contact structure disposed on and electrically connected to the first source/drain material; wherein an offset in a widthwise direction of the channel region is between the gate contact and the first source/drain contact structure, the offset is greater than zero, and the offset is not greater than an average value of a width of the channel region and a dimension of the gate contact.

18. The semiconductor structure of claim 17, further comprising a second source/drain contact structure disposed on and electrically connected to the second source/drain material, wherein the first source/drain contact structure and the second source/drain contact structure are disposed on opposite sides of the intersect region of the channel region.

19. The semiconductor structure of claim 18, wherein the first source/drain contact structure is symmetrical with the second source/drain contact structure with respect to the gate stack.

20. The semiconductor structure of claim 17, wherein a ratio between the dimension of the gate contact and a width of the gate stack is from about 2.4 to about 3.2.
